

Wine Sales Performance Analysis

A Data Analysis Project — Revenue, Regional, Product and Logistics Trends (2010–2013)

Role	Data Analyst
Tool Used	Microsoft Excel (PivotTables, Dashboard) + Python for extended analysis
Dataset	9,367 order line items · 6,429 orders · Australia, 2010–2013
Objective	Analyze wine sales performance across segments, regions, products and shipping to surface actionable

Project Overview

This project analyzes four years of wine sales data (2010–2013) for an Australian wine distributor, covering 9,367 order line items across 6,429 unique orders. The raw transactional data — spanning wine segment, region, product, packaging, freight mode and pricing — was summarized in Excel using PivotTables and a summary dashboard, then extended with deeper exploratory analysis to answer key performance questions for the business.

Business Questions Addressed

1. How has revenue trended year over year?
2. Which wine segment and region generate the most revenue?
3. Which products and varietals are the strongest performers?
4. How efficient is the shipping/freight operation, and which packaging types dominate sales?

Methodology

Data cleaning: standardized date fields and verified no missing values across the 20 source columns. Aggregation: built summary tables in Excel (PivotTables for revenue, order count, and quantity by year and state) and extended the analysis with grouped calculations for segment, region, product, sub-segment, packaging type and freight mode. Derived metrics: computed average delivery time (dispatch date minus order date) and freight cost as a percentage of revenue to assess shipping efficiency.

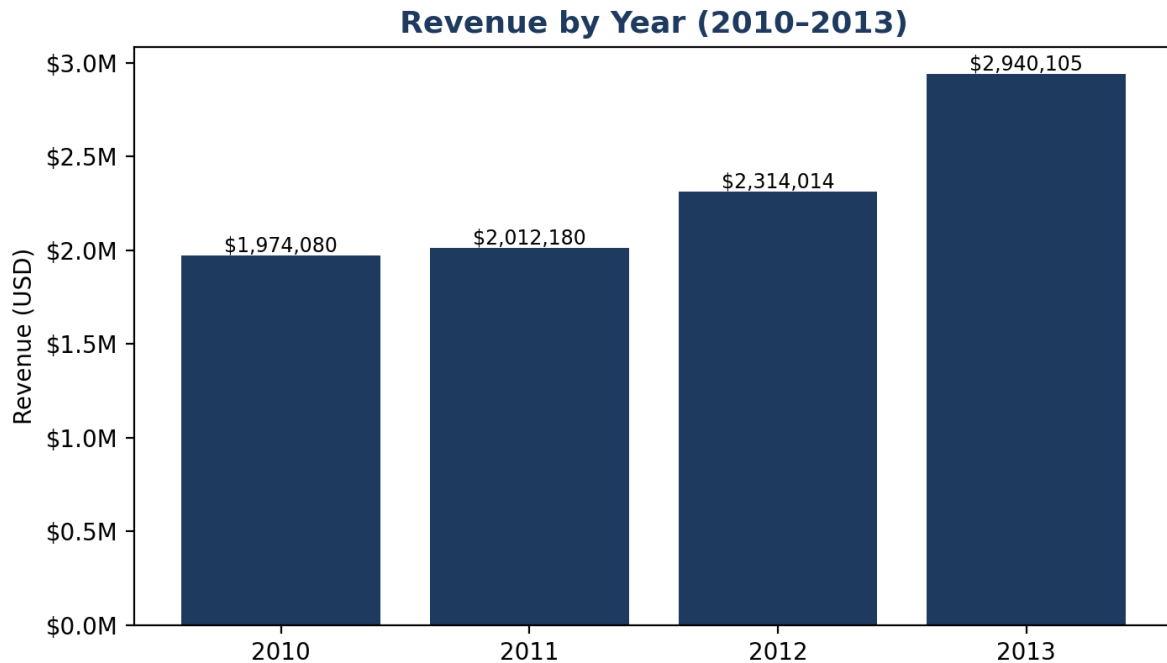
Headline Numbers

Total Revenue	Total Orders	Total Units Sold	Avg. Discount
\$9,240,379	6,429	129,284	5.0%

Key Findings

1. Revenue Trend (2010–2013)

Revenue grew every year of the analysis window, rising from \$1.97M in 2010 to \$2.94M in 2013 — a 48.9% increase over four years. Growth accelerated in 2013, up 27% on 2012, the largest single-year jump in the dataset.

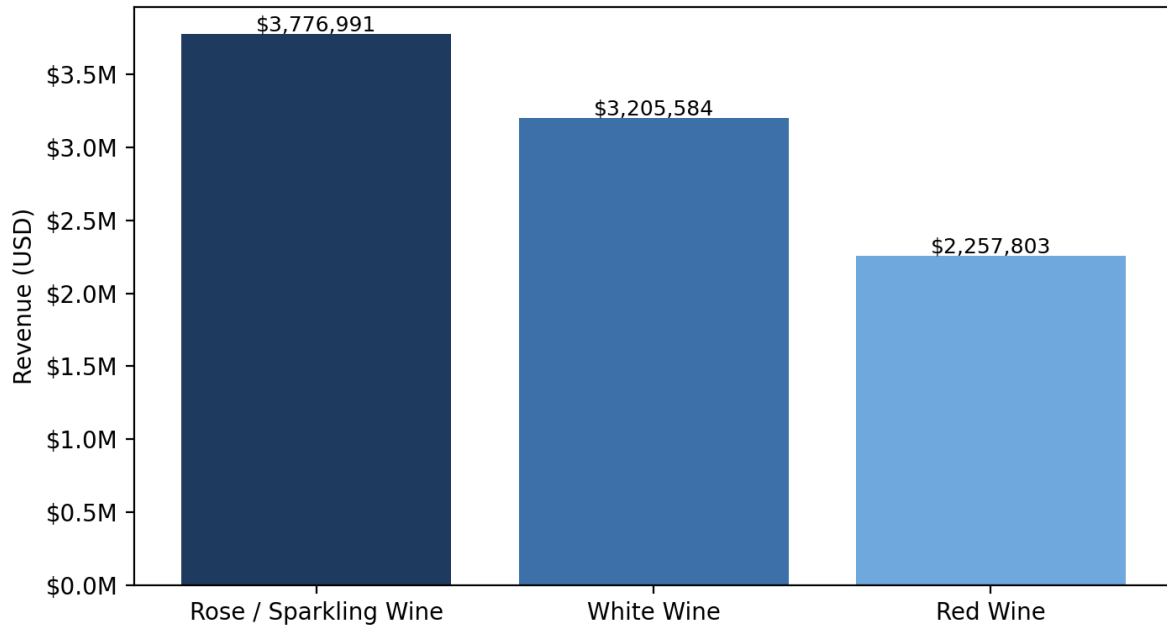


Answer: Revenue rose from \$1.97M (2010) to \$2.94M (2013), +48.9% overall

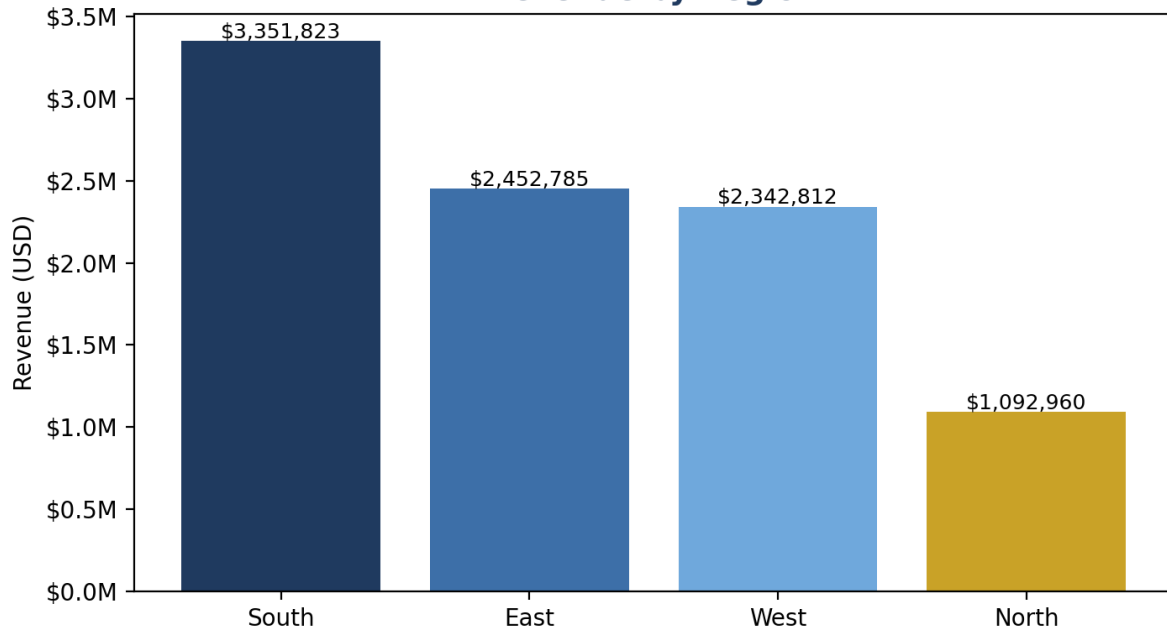
2. Best-Performing Segment & Region

Rose / Sparkling Wine is the top revenue segment at \$3.78M (40.9% of total revenue), ahead of White Wine (\$3.21M) and Red Wine (\$2.26M). Geographically, the South region leads with \$3.35M in revenue (36.3% of total), followed by East and West at roughly \$2.4M each; the North region trails substantially at \$1.09M.

Revenue by Wine Segment



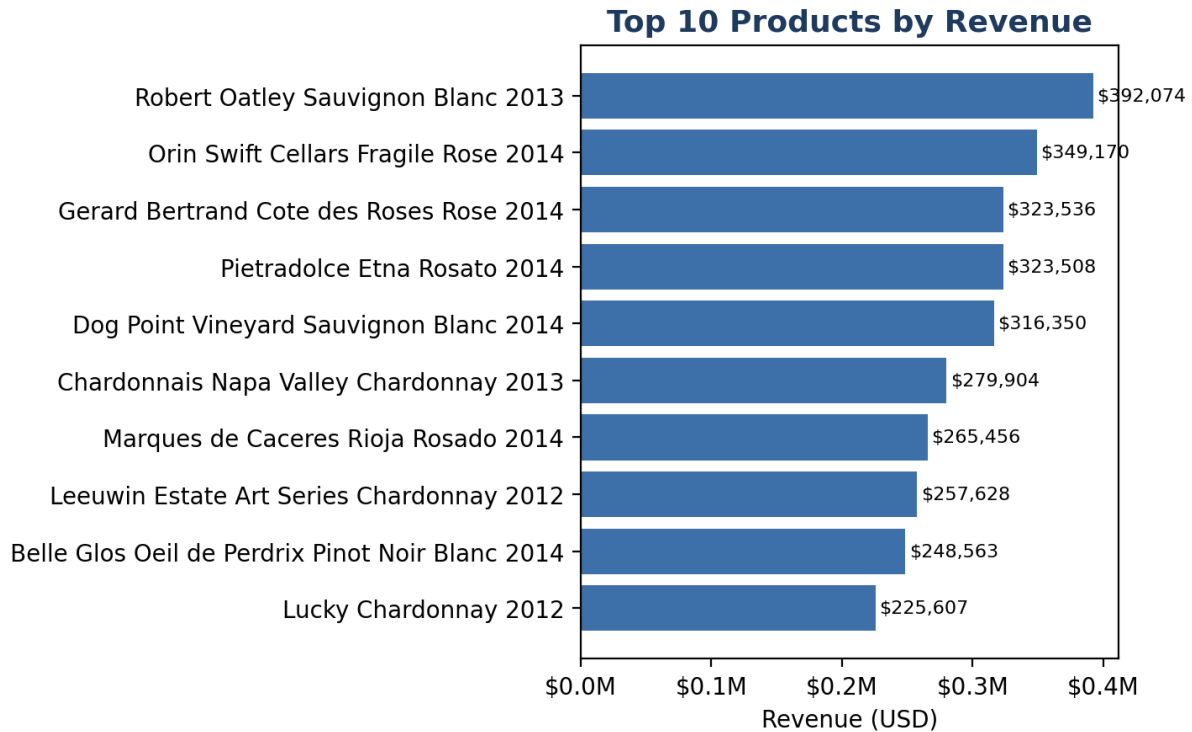
Revenue by Region



Answer: Rose / Sparkling Wine leads segments (\$3.78M); South leads regions (\$3.35M)

3. Top Products & Varietals

The single best-selling product is Robert Oatley Sauvignon Blanc 2013, generating \$392,074 in revenue, followed closely by Orin Swift Cellars Fragile Rose 2014 (\$349,170) and Gerard Bertrand Cote des Roses Rose 2014 (\$323,536). At the varietal level, Pinot Gris is the strongest performer overall (\$1.06M), with Reisling, Sauvignon Blanc, Champagne and Chardonnay rounding out the top five.

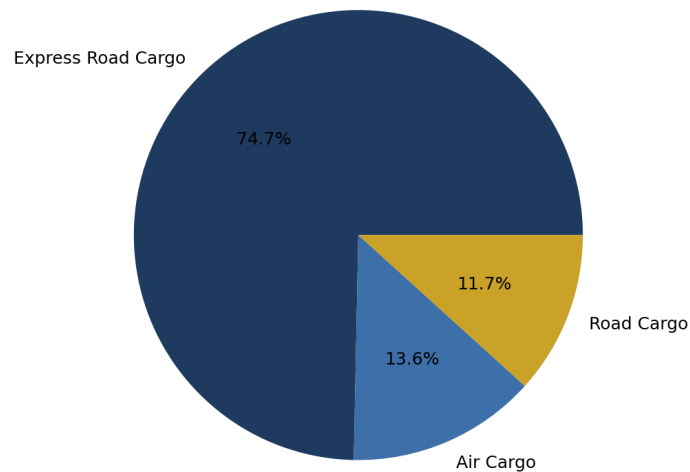


Answer: Top product — Robert Oatley Sauvignon Blanc 2013 (\$392K); top varietal — Pinot Gris (\$1.06M)

4. Shipping & Packaging Efficiency

Express Road Cargo handles the large majority of orders (74.7%), with Air Cargo (13.6%) and standard Road Cargo (11.7%) making up the rest. Express Road Cargo is also the most cost-efficient mode, with freight expense at just 1.13% of revenue, versus 1.52% for Air Cargo. Average delivery time across all orders is about 2 days. Large Crate is the dominant packaging format, accounting for \$3.30M in revenue (35.7% of total), ahead of Extra Large Crate (\$2.13M).

Order Share by Freight Mode



Answer: Express Road Cargo dominates (74.7% of orders) and is most cost-efficient (1.13% of revenue)

Regional & State Breakdown

New South Wales is the top revenue-generating state (\$2.56M), followed by Western Australia (\$2.18M). The remaining six states/territories each contribute between \$0.5M and \$1.0M.

